

Shreyansh Modi

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Education

St. Teresa School	2010 – 2024
Indian Institute of Technology, Roorkee B.Tech in Electrical Engineering	2024 – 2028

Experience

Machine Learning Engineer Intern <i>Infigro.ai</i>	July 2025 – Nov 2025
- Built a Data Analysis Agent capable of interpreting Bar Charts and Trendline Data for automated insights generation. - Worked extensively on Data Interpretation and Analysis pipelines to extract meaningful patterns from visual data representations.	
Research Intern <i>Electrical Engineering Department, IIT Roorkee</i>	Dec 2024 – Jan 2025
- Research focused on the intersection of Control Systems and Cybersecurity, advanced control strategies to enhance the security and resilience of cyber-physical systems. - Worked on Arduino, Matlab to simulate various types of attacks on cyber-physical systems. - Devised new strategies to mitigate different kinds of cyber attacks on cyber-physical systems.	

Publications

Guidance for Low-Level Perceptual Editing in Unconditional Diffusion Models	
S. Modi, A. Tomar, A. Aggarwal arXiv Code	CVPR'26 GCV Workshop
Developed a method enabling semantic control over unconditional diffusion models (DDPMs) by extracting concept-specific direction vectors from the model's internal h-space and applying them as targeted perturbations during the reverse generative process.	

Projects

HireSense	GitHub Code
- Developed an NLP-based Candidate Selection Model designed to parse resumes and job descriptions using fine-tuned NER models. - Built resume and job description parsing modules based on NLP/NER techniques and calculated similarity scores from extracted feature vectors. - Tools Used: Python, Natural Language Processing, fine-tuned NER, Git.	
InVisionDX	GitHub Code
- Developed Web2-integrated disease prediction models based on Computer Vision and Convolutional Neural Networks, achieving over 95% accuracy. - Incorporated several CNN-based models for predicting diseases such as COVID-19, Pneumonia, Tuberculosis, Lung Cancer, and Alzheimer's disease from visual data. - Tools Used: Python, PyTorch, Flask, Computer Vision, CNNs, HTML, CSS, JavaScript.	
ERP API Automated Framework	GitHub Code
- Developed an ERP Management System designed for high-performance handling and visualization of ERP-style datasets. - Developed various data analysis modules to extract valuable information from datasets and visualize them in a comprehensive manner. - Tools Used: Python, Flask, JavaScript, HTML, CSS, Git.	
Adobe Problem Statement – Photo Editing App	4th Place <i>InterIIT Tech Meet 14.0</i>
- Developed a photo editing application featuring automated editing capabilities powered by Computer Vision techniques. - Implemented intelligent image processing pipelines enabling context-aware, automated photo enhancements and edits without manual intervention. - Secured 4th position among all IITs in the Adobe problem statement at InterIIT Tech Meet 14.0. - Tools Used: Python, Computer Vision, image processing, PyTorch.	

Key Courses Taken

Mathematics: Linear Algebra, Probability and Statistics, Numerical Methods, Optimization, Calculus
Other Relevant Courses: C++ Programming, Machine Learning, Deep Learning

Positions of Responsibility

Core Member: Data Science Group , IIT Roorkee	Jan 2025 - Present
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